

**Project Title**  
**Sales Data Analysis System**

A complete end-to-end Data Analysis project using:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- MySQL
- Jupyter Notebook
- Power BI

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**Project Objective**

Analyze company sales data to:

- find top-selling products
- analyze monthly revenue
- identify profitable regions
- analyze customer behavior
- create dashboards
- generate business insights

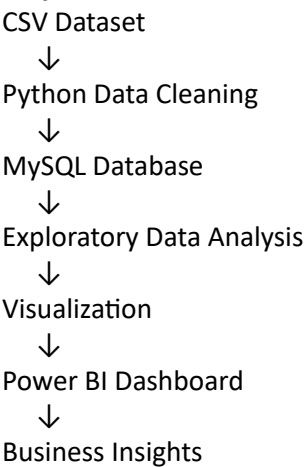
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**Technologies Used**

| Technology                   | Purpose              |
|------------------------------|----------------------|
| Python                       | Data analysis        |
| Pandas                       | Data cleaning        |
| NumPy                        | Numerical operations |
| Matplotlib                   | Visualization        |
| Seaborn                      | Advanced charts      |
| MySQL                        | Database storage     |
| Power BI                     | Dashboard creation   |
| Jupyter Notebook Development |                      |

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**Project Architecture**



### Step 1 — Install Required Libraries

```
pip install pandas numpy matplotlib seaborn mysql-connector-python jupyter
```

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### Step 2 — Dataset

Use dataset columns:

| Column        | Description      |
|---------------|------------------|
| Order_ID      | Order Number     |
| Order_Date    | Date             |
| Customer_Name | Customer         |
| Product       | Product Name     |
| Category      | Product Category |
| Region        | Sales Region     |
| Quantity      | Units Sold       |
| Price         | Product Price    |
| Sales         | Total Sales      |
| Profit        | Profit           |

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### Dataset

sales\_data.csv

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### Step 3 — Create MySQL Database

#### Create Database

```
CREATE DATABASE sales_project;  
USE sales_project;
```

---

#### Create Table

```
CREATE TABLE sales (  
    order_id INT PRIMARY KEY,  
    order_date DATE,  
    customer_name VARCHAR(100),  
    product VARCHAR(100),  
    category VARCHAR(50),  
    region VARCHAR(50),  
    quantity INT,  
    price DECIMAL(10,2),  
    sales DECIMAL(10,2),  
    profit DECIMAL(10,2)  
);
```

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### Step 4 — Load Data Using Python

```
import pandas as pd  
import mysql.connector
```

```
# Load CSV
```

```
df = pd.read_csv('sales_data.csv')
```

```
# MySQL Connection
```

```

conn = mysql.connector.connect(
    host='localhost',
    user='root',
    password='your_password',
    database='sales_project'
)

cursor = conn.cursor()

# Insert Data

for index, row in df.iterrows():
    query = """
    INSERT INTO sales
    VALUES (%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s,%s)
    """

    values = (
        int(row['Order_ID']),
        row['Order_Date'],
        row['Customer_Name'],
        row['Product'],
        row['Category'],
        row['Region'],
        int(row['Quantity']),
        float(row['Price']),
        float(row['Sales']),
        float(row['Profit'])
    )

    cursor.execute(query, values)

conn.commit()
print('Data Inserted Successfully')

```

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## Step 5 — Data Analysis Using Pandas

### Load Dataset

```
import pandas as pd
```

```
# Read CSV
```

```
df = pd.read_csv('sales_data.csv')
```

```
print(df.head())
```

---

## Step 6 — Basic Data Exploration

### Check Shape

```
print(df.shape)
```

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### Check Columns

```
print(df.columns)
```

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### Check Missing Values

```
print(df.isnull().sum())
```

---

### Dataset Information

```
print(df.info())
```

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### Statistical Summary

```
print(df.describe())
```

---

## Step 7 — Data Cleaning

### Remove Null Values

```
df.dropna(inplace=True)
```

---

### Remove Duplicate Records

```
df.drop_duplicates(inplace=True)
```

---

### Convert Date Column

```
df['Order_Date'] = pd.to_datetime(df['Order_Date'])
```

---

## Step 8 — Data Analysis Questions

### 1. Total Sales

```
print(df['Sales'].sum())
```

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### 2. Total Profit

```
print(df['Profit'].sum())
```

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### 3. Top Selling Products

```
print(df.groupby('Product')['Sales'].sum().sort_values(ascending=False))
```

---

### 4. Region-wise Sales

```
print(df.groupby('Region')['Sales'].sum())
```

---

### 5. Category-wise Profit

```
print(df.groupby('Category')['Profit'].sum())
```

---

### 6. Monthly Sales Analysis

```
df['Month'] = df['Order_Date'].dt.month
```

```
print(df.groupby('Month')['Sales'].sum())
```

---

### 7. Best Customer

```
print(df.groupby('Customer_Name')['Sales'].sum().sort_values(ascending=False).head(5))
```

---

## Step 9 — Data Visualization

### Import Libraries

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

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### 1. Sales by Region

```
region_sales = df.groupby('Region')['Sales'].sum()
```

```
region_sales.plot(kind='bar')
```

```
plt.title('Region Wise Sales')
```

```
plt.xlabel('Region')
```

```
plt.ylabel('Sales')
```

```
plt.show()
```

---

### 2. Monthly Sales Trend

```
monthly_sales = df.groupby('Month')['Sales'].sum()
```

```
plt.plot(monthly_sales)
```

```
plt.title('Monthly Sales Trend')
```

```
plt.xlabel('Month')
```

```
plt.ylabel('Sales')
```

```
plt.show()
```

---

### 3. Profit Distribution

```
sns.histplot(df['Profit'])
```

```
plt.title('Profit Distribution')
```

```
plt.show()
```

---

### 4. Correlation Heatmap

```
sns.heatmap(df.corr(numeric_only=True), annot=True)
```

```
plt.show()
```

---

## Step 10 — Advanced Analysis

### Top 5 Products by Profit

```
profit_products = df.groupby('Product')['Profit'].sum()
```

```
print(profit_products.sort_values(ascending=False).head())
```

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### Average Order Value

```
print(df['Sales'].mean())
```

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### Highest Sales Region

```
print(df.groupby('Region')['Sales'].sum().idxmax())
```

---

## Step 11 — SQL Analysis Queries

### Total Revenue

```
SELECT SUM(sales) AS total_revenue
FROM sales;
```

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### Top Products

```
SELECT product, SUM(sales) AS total_sales
FROM sales
GROUP BY product
ORDER BY total_sales DESC;
```

---

### Region-wise Profit

```
SELECT region, SUM(profit) AS total_profit
FROM sales
GROUP BY region;
```

---

### Best Customers

```
SELECT customer_name, SUM(sales) AS total_purchase
FROM sales
GROUP BY customer_name
ORDER BY total_purchase DESC
LIMIT 5;
```

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### Step 12 — Connect MySQL with Python

```
query = 'SELECT * FROM sales'
```

```
sales_df = pd.read_sql(query, conn)
```

```
print(sales_df.head())
```

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### Step 13 — Power BI Dashboard

Create Dashboard Visuals:

| Visual     | Purpose             |
|------------|---------------------|
| KPI Cards  | Total Sales, Profit |
| Line Chart | Monthly Trend       |
| Pie Chart  | Category Share      |
| Bar Chart  | Region Sales        |
| Table      | Customer Details    |
| Slicer     | Filter by Region    |

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### Dashboard KPIs

Important KPIs:

- Total Revenue
  - Total Profit
  - Profit Margin
  - Top Region
  - Top Product
  - Monthly Growth
- 

### Step 14 — Business Insights

Example insights:

1. Electronics category generated highest revenue.
2. Delhi region contributed maximum sales.
3. Profit margins are low in accessories.

4. Monthly sales increased in festive season.
5. Laptop sales dominate total revenue.

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### Step 15 — Project Folder Structure

Sales\_Data\_Analysis\_Project/

```
|
|  └── data/
|      └── sales_data.csv
|
|  └── notebooks/
|      └── analysis.ipynb
|
|  └── sql/
|      └── sales_queries.sql
|
|  └── dashboard/
|      └── sales_dashboard.pbix
|
|  └── scripts/
|      └── data_loader.py
|
|  └── README.md
```

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### Step 16 — Resume Project Description

#### Resume Line

Developed an end-to-end Sales Data Analysis project using Python, Pandas, MySQL, and Power BI to analyze revenue, profit trends, customer behavior, and regional sales performance.

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### Interview Questions Based on This Project

1. What is data cleaning?
2. Why use Pandas?
3. Difference between NumPy and Pandas?
4. What are KPIs?
5. Explain groupby() in Pandas.
6. Why use heatmaps?
7. Difference between INNER JOIN and LEFT JOIN?
8. How did you handle missing values?
9. Explain normalization.
10. What is EDA?

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### Real Industry Improvements

Advanced versions can include:

- Machine Learning prediction
  - Customer segmentation
  - Recommendation system
  - Real-time dashboard
  - Cloud deployment
  - ETL pipelines
  - Data warehouse
  - Apache Spark
-

### Mini Tasks for Practice

1. Add discount column.
2. Create yearly sales report.
3. Find least profitable products.
4. Add customer segmentation.
5. Create Power BI filters.
6. Build profit prediction model.

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### Final Understanding

This project teaches:

- Data Cleaning
- Data Visualization
- SQL Queries
- Dashboarding
- Business Intelligence
- Database Integration
- Analytical Thinking

This is the type of project commonly used by:

- Data Analysts
- Business Analysts
- BI Developers
- Junior Data Scientists

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### End of Project